The Electron Field Equation of Temporal Undulatory Theory (TauTimp): Relation to the Schrödinger Radial Equation and the Structure of the Hydrogen Atom A radial wave equation with a curvature factor, its Taylor comparison with the Schrödinger equation, and the standard hydrogen-atom results (spectrum, fine structure, Lamb shift, 21 cm line, Zeeman effect, orbitals) with explicit status labels Viorel-Mihai Popescu 2 October 2026

Abstract

Temporal Undulatory Theory (TauTimp) represents the electron by a real field $f(r,t)$ obeying a wave equation in which the radial operator carries a curvature factor R/r^2, with $R = 4.2355 \times 10^{22}$ m [P]. This note examines whether that equation reduces, at atomic scale, to the Schrödinger radial equation of hydrogen. Three results are stated. (1) The equation as given in the working specification [6], compared coefficient by coefficient with the Schrödinger equation through a Taylor expansion at $r = 0$, differs from it at order r^1 (for $l > 0$) or r^2 ($l = 0$) [M]. (2) Three algebraic modifications (sign of the Coulomb term, addition of the bare energy term, localisation of the centrifugal term) turn that equation into the Schrödinger equation plus one extra term, the "cosmological tail" T, which is part of the specification and is a postulate [P]. The Taylor coincidence through the orders listed in Section 4.5 is therefore a consequence of the construction and not an independent test of TauTimp. A fourth modification (removal of a factor r/R) is needed if one starts from the derivation of Section 3.3 instead of from the specification. (3) The only quantitative content of the construction is the size of the tail under the reading T'. As printed, the tail has dimension m^0 in a bracket whose other terms have dimension m^{-2}, so its ratio to the Coulomb term has the unit m^2 and its value depends on the unit of length. For the dimensionally consistent reading $T' = (r/R)^4(2m_eE/\hbar^2 - \kappa^2)$ the ratio is dimensionless, equal to 1.5×10^{-131}, 4.6×10^{-128} and 1.9×10^{-121} at $r = 0.2,\ 1,\ 21\ a_0$ [M], and the first-order shift of the $1s$ level is 2.8×10^{-125} eV [M]; $\langle T' \rangle$ for other states scales as $\langle r^4 \rangle$ and is not tabulated here. For the printed form, dividing by L^2 for the missing length L gives at most 2.2×10^{-51} for any $L \geq \bar{\lambda}_C$ [M, conditional]. The remaining sections present the standard results for hydrogen (Bohr spectrum, Dirac fine structure, Lamb shift, 21 cm line, normal Zeeman effect, orbitals) as established physics [E], recomputed from the stated constants; changes relative to the unpublished draft of 26 August 2026 are listed in Appendix B. A probability-density wave equation is retained as a hypothesis [H], with its dimensional and numerical errors marked. A discriminating test of TauTimp would have to be made at $r \sim R$ and is not attempted here.

Keywords: TauTimp, radial wave equation, Schrödinger equation, Taylor expansion, hydrogen atom, fine structure, hyperfine structure

1. Introduction and status of claims 1.1 Labels

Every substantive statement carries one of the following labels. An unlabelled statement is a definition, a description of the model, or narrative.

Label	Meaning
[E]	Established: standard mathematics or physics, textbook or measured
[M]	Mathematical identity, or explicit arithmetic from the stated inputs (the calculation is shown)
[P]	Postulate: assumed in TauTimp, not derived here
[H]	Hypothesis: proposed, with a motivation but without derivation or evidence in this note

1.2 Context and question

TauTimp postulates that the electron is an excitation of a real spatio-temporal field $f(r,t)$ propagating on a background with cosmological scale R [P]. The field equation contains, besides the Coulomb and centrifugal terms, the radial operator $(R/r^2)\ \partial_r^2(rf)$ [P]. The question examined here is under which precise conditions, and with which status, this equation reduces at $r \sim a_0 \ll R$ to the Schrödinger radial equation of hydrogen.

1.3 The three tasks of the working specification

The working specification [6] states the field equation (Section 3.1) and requests: (a) a comparison of the ODE model with the Bohr/Schrödinger model for the spectral lines, the Lamb shift and the hyperfine structure; (b) a comparison of the Taylor expansions of the Schrödinger and undulatory radial equations, with equal free term r^0 and equal values for $r \lesssim 21\ a_0$; (c) detailed explanations, with plots, of the two equations at small r. Sections 3–4 treat (b) and (c); Sections 5–11 treat (a) and the standard atomic results. The specification also asks for the energy E in eV; all formulas below are evaluated in SI units (energies in J), and the conversion $1\ \mathrm{eV} = 1.602176634 \times 10^{-19}$ J is implicit where energies are quoted in eV.

1.4 What this note is not

(i) It does not derive the TauTimp equation from a more fundamental principle. (ii) It does not derive the Lamb shift. (iii) It does not claim that TauTimp is needed to explain hydrogen. (iv) It does not test TauTimp, because the only place where TauTimp differs from Schrödinger is at $r \sim R$.

2. Notation and constants

Table 1 — Constants (CODATA 2018 [5]).

Symbol	Value	Description
$\hbar$	$1.054571817 \times 10^{-34}$ J·s	reduced Planck constant

Symbol	Value	Description
$h = 2\pi\hbar$	$6.62607015 \times 10^{-34}$ J·s	Planck constant
m_e	$9.1093837015 \times 10^{-31}$ kg	electron mass
c	2.99792458×10^{8} m/s	speed of light
e	$1.602176634 \times 10^{-19}$ C	elementary charge
ε_0	$8.8541878128 \times 10^{-12}$ F/m	vacuum permittivity
α	$7.2973525693 \times 10^{-3}$	fine-structure constant
a_0	$5.29177210903 \times 10^{-11}$ m	Bohr radius
$\bar{\lambda}_C = \hbar/(m_e c) = \alpha a_0$	$3.8615927 \times 10^{-13}$ m	reduced Compton wavelength
R	4.2355×10^{22} m	TauTimp cosmological radius [P]
R_y	13.605693 eV	Rydberg energy
hc	1239.84193 eV·nm	
m_p	$1.67262192 \times 10^{-27}$ kg	proton mass
g_p	5.5856947	proton g-factor
μ_B	$9.2740101 \times 10^{-24}$ J/T	Bohr magneton
μ_N	$5.0507837 \times 10^{-27}$ J/T	nuclear magneton
μ_0	$1.25663706 \times 10^{-6}$ N/A²	vacuum permeability
$m_e c^2$	510998.95 eV	$= 9.1093837015 \times 10^{-31} \times (2.99792458 \times 10^{8})^2 / 1.602176634 \times 10^{-19}$ [M]

Derived quantities used below [M]:

- $\kappa^2 \equiv 2m_e^2 c^2/\hbar^2 = 2/\bar{\lambda}_C^2$. The specification does not define the symbol k that appears in its tail; κ^2 is introduced here as a dimensionally consistent reading (the expression $2m_e c^2/\hbar^2$ has dimension $1/(\mathrm{kg\,m^2})$, whereas the neighbouring term $2m_e E/\hbar^2$ has dimension $\mathrm{m^{-2}}$; E4). With $1/\bar{\lambda}_C = 2.589605 \times 10^{12}$ $\mathrm{m^{-1}}$, $\kappa^2 = 1.34121 \times 10^{25}$ $\mathrm{m^{-2}}$.
- $1/R^2 = 1/(4.2355 \times 10^{22})^2 = 5.574 \times 10^{-46}$ $\mathrm{m^{-2}}$ ($R^2 = 1.79394 \times 10^{45}$ $\mathrm{m^2}$).
- $a_0/R = 1.2494 \times 10^{-33}$.
- The local wavenumber of Section 3.2 is called $q(r)$, the time factor $\tau(t)$, the reduced mass m_r, and the probability density of Section 12 $\varrho(r,t)$, so that no symbol is used twice (θ is reserved for the polar angle). R is the cosmological radius, R_{nl} the hydrogen radial function and R_y the Rydberg energy (standard notation).

Sign convention. Equation (5) below uses the physical signs (+ Coulomb attraction, − centrifugal term in the bracket). The working specification prints the opposite signs; the difference is stated in Section 3.1.

3. The TauTimp electron equation

3.1 The equation of the working specification

The working specification [6] gives the electron equation, with $f(r,t) = g(r)\,\tau(t)$ (written $g(r)u(t)$ in the specification) and the boundary condition $g(R) = 0$, in the form

$$\frac{2m_e}{\hbar^2}\left(E - \frac{e^2}{4\pi\varepsilon_0 r} + \frac{l(l+1)\hbar^2}{2m_e r^2}\right) f + \frac{R}{r^2}\frac{\partial^2 (rf)}{\partial r^2} = \frac{1}{V^2}\frac{\partial^2 f}{\partial t^2} \qquad (1)$$

where V is the electron velocity and l the orbital quantum number [P]. In the specification the Coulomb term is printed without the factor $1/r$; this is read here as a typographical omission and r is restored. The boundary condition $g(R) = 0$ is a postulate of the theory [P] and is not used in the results of this note. Equation (1) has the Coulomb term with sign − and the centrifugal term with sign +; the equation used in this note is (5), obtained from the energy balance below, which differs from (1) by the signs of these two terms [P] (E3).

3.2 Energy balance and curvature operator

For a hydrogenic electron the semi-classical energy balance is [E]

$$E = 1/2\, m_e V^2 - \frac{e^2}{4\pi\varepsilon_0 r} + \frac{l(l+1)\hbar^2}{2m_e r^2}, \qquad (2)$$

so that

$$q^2(r) \equiv \frac{m_e^2 V^2}{\hbar^2} = \frac{2m_e}{\hbar^2}\left[E + \frac{e^2}{4\pi\varepsilon_0 r} - \frac{l(l+1)\hbar^2}{2m_e r^2}\right]. \qquad (3)$$

The curvature-corrected d'Alembert operator of TauTimp is postulated as [P]

$$\Box f = \frac{R}{r^2}\frac{\partial^2 (rf)}{\partial r^2} - \frac{1}{V^2}\frac{\partial^2 f}{\partial t^2} = 0. \qquad (4)$$

For f independent of the angles, $(rf)'' = r\,(f'' + 2f'/r)$, so $(R/r^2)(rf)'' = (R/r)\,\Delta f$ [M]. Hence (4) is the ordinary wave equation with the Laplacian multiplied by R/r; at $r = R$ it coincides with the ordinary d'Alembertian. At $r = a_0$ the factor is $R/a_0 = 8.00 \times 10^{32}$ [M] (E8). Combining (3) and (4) in the manner of the specification, with the physical signs, gives

$$\frac{2m_e}{\hbar^2}\left[E+\frac{e^2}{4\pi\varepsilon_0 r}-\frac{l(l+1)\hbar^2}{2m_e r^2}\right]f+\frac{R}{r^2}\frac{\partial^2(rf)}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 f}{\partial t^2}. \qquad (5)$$

3.3 Separation of variables

Take $\tau(t)=\cos\omega t$ with $E=\hbar\omega$. Substituting in (5) gives $(2m_e/\hbar^2)[\dots]\,g+(R/r^2)(rg)''+(\omega^2/V^2)\,g=0$. The step by which the last term is absorbed into $2m_eE/\hbar^2$ for $V^2=2E/m_e$ is not shown: for bound states $E<0$, so $V^2=2E/m_e<0$; and with $V^2=2E/m_e$ and $E=\hbar\omega$ one obtains $\omega^2/V^2=m_e\omega/(2\hbar)$, the factor $1/4$ of $2m_eE/\hbar^2=2m_e\omega/\hbar$ [M]. The radial equation

$$\frac{R}{r^2}\frac{d^2(rg)}{dr^2}+g\left[\frac{2m_eE}{\hbar^2}+\frac{2}{a_0 r}-\frac{l(l+1)}{r^2}\right]=0, \qquad (6)$$

with $2m_e e^2/(4\pi\varepsilon_0\hbar^2)=2/a_0$ [E], is therefore adopted as a postulate [P] and not derived from (5) (E14).

3.4 Spin extension

To describe fine and hyperfine structure the equation is promoted to act on a spinor $\Psi=f(r,t)\chi_s$, with spin-orbit coupling $\xi(r)\vec{L}\cdot\vec{S}$ and the Fermi contact term $V_{HF}(r)\,\vec{S}_e\cdot\vec{I}_p$ added to the bracket of (5) [P]:

$$\frac{2m_e}{\hbar^2}\left[E+\frac{e^2}{4\pi\varepsilon_0 r}-\frac{l(l+1)\hbar^2}{2m_e r^2}+\xi(r)\langle\vec{L}\cdot\vec{S}\rangle+V_{HF}(r)\langle\vec{S}_e\cdot\vec{I}_p\rangle\right]f+\frac{R}{r^2}\frac{\partial^2(rf)}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 f}{\partial t^2}, \qquad (7)$$

with $\xi(r)=\hbar\alpha/(2m_e^2 c\,r^3)$ (Coulomb) [E] and $V_{HF}(r)=\frac{4\mu_0}{3\hbar^2}\,g_p\mu_B\mu_N\,\delta^3(\vec{r})$ [E, 1, 2] (E10). Equation (7) contains the spin-orbit term and the Fermi term; it contains neither the Darwin term nor the p^4 correction (Sections 6.2–6.3). Fine structure is obtained from the three terms by standard first-order perturbation theory (Section 7).

4. Comparison with the Schrödinger radial equation 4.1 The two forms

With $u(r)=r\,g(r)$ the Schrödinger radial equation of hydrogen is [E]

$$\mathcal{L}_S[u]\equiv u''+u\,B_S=0, \qquad B_S=\frac{2m_eE}{\hbar^2}+\frac{2}{a_0 r}-\frac{l(l+1)}{r^2}. \qquad (8)$$

Multiplying (6) by r^2/R and using $(rg)''=u''$ gives, exactly [M],

$$u''+\frac{r}{R}\,u\,B_S=0. \qquad (9)$$

The factor $r/R=1.25\times10^{-33}$ at $r=a_0$ multiplies the whole potential, so (9) is **not** the Schrödinger equation: at atomic scale the potential and the energy would be suppressed by thirty-three orders of magnitude.

The working specification [6] requests the Taylor comparison with a different expression, in terms of g:

$$2g'+r\,g''+r\,g\left[\frac{r^4}{R^2}\left(\frac{2m_eE}{\hbar^2}-\kappa^2\right)-\frac{2}{a_0 r}-\frac{l(l+1)}{R}\right]=0. \qquad (10)$$

Since $2g'+rg''=u''$ and $rg=u$, (10) is $u''+u\,B_{\text{raw}}=0$ with B_{raw} the bracket of (10) [M] (the specification's unnamed constant k is written κ, Section 2). Equation (10) has no factor r/R in front of the potential, so it is not obtained from (6) or (9) by any algebra shown here. Its relation to (6) is a postulate of the specification [P]; the passage from (9) to the Schrödinger form requires removing the factor r/R (modification (iv) below).

Two-dimensional remarks on (10) [M]. (a) The last term, $l(l+1)/R$, has dimension m^{-1} while the other terms of the bracket have dimension m^{-2} (modification (iii) removes it; E17). (b) The first term, $r^4(2m_eE/\hbar^2-\kappa^2)/R^2$, has dimension m^0: r^4/R^2 is a length squared and the parenthesis is m^{-2}. This is the "cosmological tail"; it belongs to the specification [P] and its dimension is discussed in Section 4.6.

4.2 Method of the Taylor comparison

Write $u(r)=\sum_{n\geq1} g_{n-1}\,r^n$, i.e. $u_0=0$ and $u_n=g_{n-1}$ (from $u=rg$), with $g_0, g_1, \dots$ arbitrary. Each expression is multiplied by r^2 so that it is a power series, and coefficients are compared at the same physical power of r. Comparison at the same index of different multipliers (r^2 against r) gives false divergences and is not used. No assumption beyond $u=rg$ is made on the g_n.

4.3 The specification equation (10) against Schrödinger

Define $D(r)=r^2\,(\mathcal{L}_S[u]-\mathcal{L}_{\text{raw}}[u])=r^2u\,(B_S-B_{\text{raw}})$ with $\mathcal{L}_{\text{raw}}[u]\equiv u''+uB_{\text{raw}}$. Then

$$B_S-B_{\text{raw}}=\frac{2m_eE}{\hbar^2}+\frac{4}{a_0 r}-\frac{l(l+1)}{r^2}+\frac{l(l+1)}{R}-\frac{r^4}{R^2}\left(\frac{2m_eE}{\hbar^2}-\kappa^2\right),$$

and expanding $u=g_0r+g_1r^2+\cdots$ the coefficients of D are [M]

$$[r^0]D=0, \qquad [r^1]D=-g_0\,l(l+1), \qquad [r^2]D=-g_1\,l(l+1)+\frac{4g_0}{a_0}. \qquad (11)$$

The free term r^0 is zero because $u(0)=0$, a consequence of $u=rg$ alone; this satisfies the condition "r^0 equal" of the specification trivially, before and after any modification. The first non-zero coefficient is at r^1 for $l>0$ and at r^2 for $l=0$ (where it equals $4g_0/a_0$) (E5). Hence the expression (10), taken literally, does not reproduce the Schrödinger equation at small r [M]. The structural reasons are visible in B_S-B_{raw}: the sign of the Coulomb term, the absence of the bare energy term $2m_eE/\hbar^2$ (which appears in (10) only inside the tail), and the global $(1/R)$ instead of local $(1/r^2)$ centrifugal term.

4.4 The modifications, and what they imply

#	Modification	Effect on (10)
(i)	Coulomb sign	$-2/(a_0 r) \to +2/(a_0 r)$
(ii)	add bare energy term	$+2m_e E/\hbar^2$ in the bracket
(iii)	localise centrifugal term	$-l(l+1)/R \to -l(l+1)/r^2$
(iv)	remove the factor r/R of (9)	needed only on the route from the derivation (6)

Applying (i)–(iii) to (10) gives

$$u'' + u\left[\frac{2m_e E}{\hbar^2} + \frac{2}{a_0 r} - \frac{l(l+1)}{r^2} + T(r)\right] = 0, \qquad T(r) = \frac{r^4}{R^2}\left(\frac{2m_e E}{\hbar^2} - \kappa^2\right). \qquad (12)$$

By inspection [M], (12) is identical to (8) with the extra term $u\,T(r)$: $\mathcal{L}_{\mathrm{mod}}[u] = \mathcal{L}_S[u] + T(r)\,u.$ (13)

Consequence. The modifications (i)–(iii) have the effect of converting the bracket B_{raw} into B_S; (12) is therefore the Schrödinger equation apart from T. That the Taylor coefficients of (12) and (8) coincide up to the orders of Section 4.5 is a rewriting of (13) and not an independent verification of the TauTimp equation (E1). The tail T is not added by this note; it is a term of the specification [P], and its form does not follow from (6) (after modification (iv), (9) becomes the Schrödinger equation with no tail). Under the reading T' of Section 4.6 the only quantitative content of the construction is the size of the tail. Whether (i)–(iv) are justified within TauTimp is not shown here.

4.5 Orders of agreement

With the convention of Section 4.2 (both sides multiplied by r^2), (13) gives $r^2(\mathcal{L}_S - \mathcal{L}_{\mathrm{mod}}) = -r^2 T\,u = -g_0\,\frac{2m_e E/\hbar^2 - \kappa^2}{R^2}\,r^7 + O(r^8)$ [M]. The coefficients of $r^0, \ldots, r^6$ vanish identically for arbitrary g_n, and the first non-vanishing coefficient is at r^7 (Fig. 2). In the convention in which the unmultiplied expressions are subtracted, the same statement reads: identical through r^4, first difference at r^5, $[r^5](\mathcal{L}_S[u] - \mathcal{L}_{\mathrm{mod}}[u]) = g_0\,\frac{\hbar^2\kappa^2 - 2m_e E}{R^2\hbar^2}.$ (14)

The conventions differ by the shift r^2 (E6). The raw form (10) differs at every order from r^1 (or r^2) onward, as in (11) (Fig. 2).

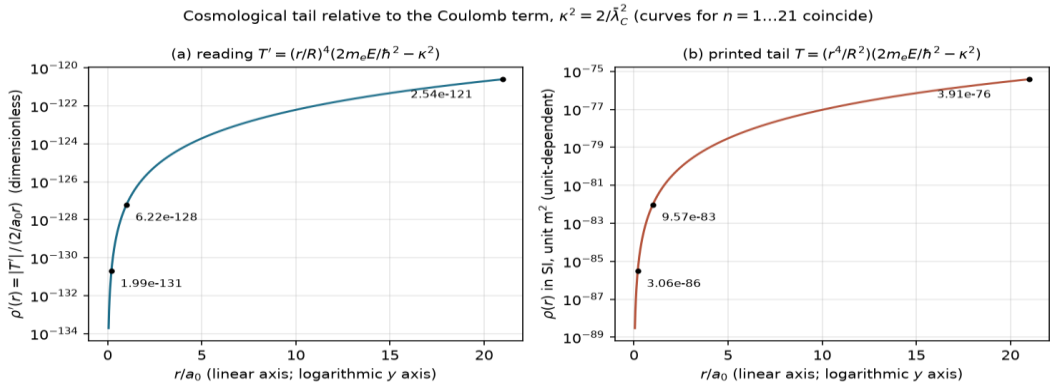


Fig. 1 — Cosmological tail relative to the Coulomb term, computed for this note, with $\kappa^2 = 2/\bar{\lambda}_C^2$; the curves for $n = 1 \ldots 21$ coincide. Linear horizontal axis $r/a_0 \in (0,21]$, logarithmic vertical axis. (a) Reading $T' = (r/R)^4(2m_eE/\hbar^2 - \kappa^2)$: $\rho'(r) = |T'|/(2/(a_0 r))$, dimensionless; marked values 1.46×10^{-131} $(0.2\,a_0)$, 4.58×10^{-128} (a_0), 1.87×10^{-121} $(21\,a_0)$. (b) Printed tail $T = (r^4/R^2)(2m_eE/\hbar^2 - \kappa^2)$: $\rho(r)$ in SI, unit m^2 (unit-dependent); marked values 2.63×10^{-86}, 8.21×10^{-83}, 3.35×10^{-76}.

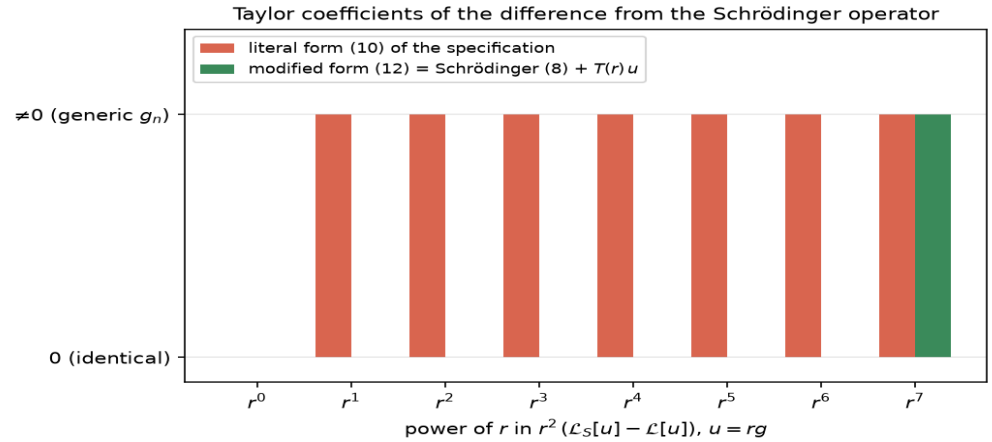


Fig. 2 — Taylor coefficients of $r^2(\mathcal{L}_S[u] - \mathcal{L}[u])$, $u = rg$, for generic g_n, computed for this note, for $r^0, \ldots, r^7$. Literal form (10) of the specification (red): zero at r^0, non-zero from r^1 to r^7. Modified form (12) = Schrödinger (8) $+ T(r)\,u$ (green): zero through r^6, non-zero only at r^7.

4.6 Size of the tail for $r \le 21\,a_0$

The tail T is present exactly, not only to Taylor order, in (12). Its ratio to the dominant small-r potential term $2/(a_0 r)$ is, to leading order in $|E|/(m_e c^2)$ [M] (Fig. 1),

$$\rho(r) = \left|\frac{T(r)}{2/(a_0 r)}\right| = \frac{r^5 \kappa^2 a_0}{2R^2} = x^5\,\frac{a_0^6 \kappa^2}{2R^2}, \qquad x = \frac{r}{a_0}. \qquad (15)$$

Calculation: $a_0^6 = (1.481847 \times 10^{-31})^2 = 2.1959 \times 10^{-62}$ m^6; $\kappa^2/2 = 1/\bar{\lambda}_C^2 = 6.7061 \times 10^{24}$ m^{-2}; $1/R^2 = 5.574 \times 10^{-46}$ m^{-2}. The product is 8.208×10^{-83}, so $\rho = 8.208 \times 10^{-83}\,x^5$. The correction from the term $2m_eE/\hbar^2$ in T is $|E|/(m_e c^2) \le 13.6/510999 = 2.7 \times 10^{-5}$ (relative) and is neglected; hence the values do not depend on n at the quoted precision.

Dimensional status of the printed tail. In (15) the numerator has dimension $\mathrm{m}^5 \cdot \mathrm{m}^{-2} \cdot \mathrm{m} = \mathrm{m}^4$ and the denominator m^2, so ρ has dimension m^2 [M]; this is why panel (b) of Fig. 1 carries the unit m^2: the printed tail is m^0 in a bracket whose other terms, including the Coulomb term, are m^{-2}. The numbers are SI values of a quantity with that unit and depend on the unit of length: at $r = 21\,a_0$, $\rho = 3.35 \times 10^{-76}$ m^2 $= 3.4 \times 10^{-46}$ fm^2, and any other length unit gives another number. No statement of negligibility follows from ρ alone.

Conditional statement [M, conditional]. If the printed tail is made homogeneous by dividing by L^2 for a missing length L, then at $r = 21\,a_0$ the ratio is $\rho(21\,a_0)/L^2 = 3.35 \times 10^{-76}$ m$^2/L^2$, which equals 1.20×10^{-55} for $L = a_0$ ($a_0^2 = 2.8003 \times 10^{-21}$ m^2), 2.25×10^{-51} for $L = \bar{\lambda}_C$ ($\bar{\lambda}_C^2 = 1.4912 \times 10^{-25}$ m^2) and 1.87×10^{-121} for $L = R$ ($R^2 = 1.7939 \times 10^{45}$ m^2). Hence for every $L \ge \bar{\lambda}_C$ the ratio at $r \le 21\,a_0$ is at most 2.2×10^{-51}. The choice of L is not made by the specification.

The dimensionally consistent reading. The reading $T'(r) = (r/R)^4(2m_eE/\hbar^2 - \kappa^2)$ has dimension m^{-2}, and its ratio to the Coulomb term is dimensionless [M]:

$$\rho'(r) = \frac{(r/R)^4 \kappa^2\, a_0 r}{2} = x^5 \left(\frac{a_0}{R}\right)^4 \frac{a_0^2}{\bar{\lambda}_C^2} = x^5\,\frac{2.4368 \times 10^{-132}}{\alpha^2} = 4.576 \times 10^{-128}\,x^5. \qquad (16)$$

(Here $(a_0/R)^4 = (1.2494 \times 10^{-33})^4 = 2.4368 \times 10^{-132}$ and $a_0/\bar{\lambda}_C = 1/\alpha$, $1/\alpha^2 = 18779$.) The choice between T and T' is a matter for the specification [P].

Table 2 — Tail-to-Coulomb ratio.

r/a_0	ρ', reading T' (dimensionless)	ρ, printed tail (SI value, unit m^2)
0.2 ($x^5 = 3.2 \times 10^{-4}$)	1.46×10^{-131}	2.63×10^{-86}
1.0	4.58×10^{-128}	8.21×10^{-83}
21 ($x^5 = 4.0841 \times 10^6$)	1.87×10^{-121}	3.35×10^{-76}

For the reading T' the ratio at $r = 21\,a_0$ lies below double-precision rounding ($\sim 10^{-16}$) by more than 100 orders of magnitude (1.87×10^{-121} against 10^{-16}: 105 orders); for the printed tail the SI value 3.35×10^{-76} is 59 orders below 10^{-16}, a number that has meaning only in metres. For the reading T', (12) is numerically indistinguishable from (8) over $r \le 21\,a_0$ at the level of the ratio (16) [M]. This is a statement about the size of an added term, not about the validity of TauTimp.

Effect on a level. For the reading T', first-order perturbation theory [E] with the equivalent potential $\Delta V = m_e c^2 (r/R)^4$ (from $\hbar^2\kappa^2/2m_e = m_e c^2$) gives for the bound-state shift $\delta E = \langle \Delta V \rangle$, which is finite. For the $1s$ state, with $\langle r^4 \rangle = 22.5\,a_0^4$ [E]: $\delta E = 22.5\,m_e c^2 (a_0/R)^4 = 22.5 \times 510999$ eV $\times\, 2.4368 \times 10^{-132} = 2.8 \times 10^{-125}$ eV, a relative shift $2.8 \times 10^{-125}/13.6057 = 2.1 \times 10^{-126}$ [M, reading T']. The shifts of other states scale as $\langle r^4 \rangle$ and are not tabulated here. For the printed form no unit-independent shift can be quoted.

4.7 Small-r behaviour

At $r = a_0$ and $l = 1$ the centrifugal term $l(l+1)/r^2 = 2/a_0^2$ and the Coulomb term $2/(a_0 r) = 2/a_0^2$ are equal [M]; the energy term has $|2m_eE/\hbar^2| = 1/(n^2 a_0^2)$ because $R_y = \hbar^2/(2m_e a_0^2)$ [M]. At the same radius the tail under the reading T' is $(a_0/R)^4\kappa^2 = 3.27 \times 10^{-107}$ m^{-2} $= 9.15 \times 10^{-128}/a_0^2$ [M], i.e. 4.6×10^{-128} of the Coulomb or the $l = 1$ centrifugal term and 9.2×10^{-128} of the $n = 1$ energy term. At $r = 21\,a_0$ the Coulomb term is $2/(21a_0^2) = 0.095/a_0^2$, the $l = 1$ centrifugal term $4.5 \times 10^{-3}/a_0^2$, the energy term $2.3 \times 10^{-3}/(a_0^2)$ for $n = 21$, and the tail $T' = 21^4 \times 3.27 \times 10^{-107}$ m^{-2} $= 1.8 \times 10^{-122}/a_0^2$; so the tail is below 10^{-118} of each of them for $r \le 21\,a_0$ and $n \le 21$ [M, reading T']. For $l = 0$ the centrifugal term is absent. The identity (13) shows that nothing else distinguishes (12) from (8). Near the origin the tail ($\propto r^4$) vanishes faster than all other terms of the bracket.

5. The hydrogen spectrum

5.1 Energy levels and spectral lines

The eigenvalues of (8) are [E]

$$E_n = -\frac{R_y}{n^2}, \qquad R_y = \frac{m_e e^4}{2(4\pi\varepsilon_0)^2\hbar^2} = 13.605693 \text{ eV}. \qquad (17)$$

After modifications (i)–(iii) the spectrum of (12) coincides with that of (8) up to the shift produced by T' (Section 4.6). For a transition $n_2 \to n_1$ ($n_2 > n_1$), $\Delta E = R_y(1/n_1^2 - 1/n_2^2)$ and $\lambda = hc/\Delta E = K/(1/n_1^2 - 1/n_2^2)$ with

$$K = \frac{hc}{R_y} = \frac{1239.84193 \text{ eV nm}}{13.605693 \text{ eV}} = 91.12670 \text{ nm}. \qquad (18)$$

All values assume infinite nuclear mass (vacuum wavelengths); they are standard results [E] recomputed from the stated constants [M]. The Balmer lines are shown in Fig. 3.

Table 3 — Spectral lines (infinite nuclear mass; $R_y = 13.605693$ eV, $hc = 1239.84193$ eV·nm).

Series	$n_2 \to n_1$	$1/n_1^2 - 1/n_2^2$	ΔE (eV)	$\lambda = K/(\ldots)$ (nm)	Region
Lyman α	$2 \to 1$	$3/4$	10.2043	121.502	UV
Lyman β	$3 \to 1$	$8/9$	12.0939	102.518	UV
Lyman γ	$4 \to 1$	$15/16$	12.7553	97.202	UV
Balmer Hα	$3 \to 2$	$5/36$	1.8897	656.112	visible (red)
Balmer Hβ	$4 \to 2$	$3/16$	2.5511	486.009	visible (blue-green)
Balmer Hγ	$5 \to 2$	$21/100$	2.8572	433.937	visible (violet)
Balmer Hδ	$6 \to 2$	$2/9$	3.0235	410.070	visible (violet)
Paschen α	$4 \to 3$	$7/144$	0.6614	1874.606	IR
Paschen β	$5 \to 3$	$16/225$	0.9675	1281.469	IR
Brackett α	$5 \to 4$	$9/400$	0.3061	4050.08	far IR

Calculations: $\lambda_{\mathrm{Ly}\alpha} = 91.12670 \times 4/3 = 121.5023$; $\lambda_{\mathrm{Ly}\beta} = 91.12670 \times 9/8 = 102.5175$; $\lambda_{\mathrm{Ly}\gamma} = 91.12670 \times 16/15 = 97.2018$; $\lambda_{\mathrm{H}\alpha} = 91.12670 \times 36/5 = 656.1122$; $\lambda_{\mathrm{H}\beta} = 91.12670 \times 16/3 = 486.0091$; $\lambda_{\mathrm{H}\gamma} = 91.12670/0.21 = 433.9367$; $\lambda_{\mathrm{H}\delta} = 91.12670 \times 9/2 = 410.0702$; $\lambda_{\mathrm{Pa}\alpha} = 91.12670 \times 144/7 = 1874.6064$; $\lambda_{\mathrm{Pa}\beta} = 91.12670 \times 225/16 = 1281.4692$; $\lambda_{\mathrm{Br}\alpha} = 91.12670 \times 400/9 = 4050.0756$. The energies are 13.605693 eV times the fractions of the third column: 10.20427, 12.09395, 12.75534, 1.88968, 2.55107, 2.85720, 3.02349, 0.66139, 0.96752, 0.30613 eV (E9).

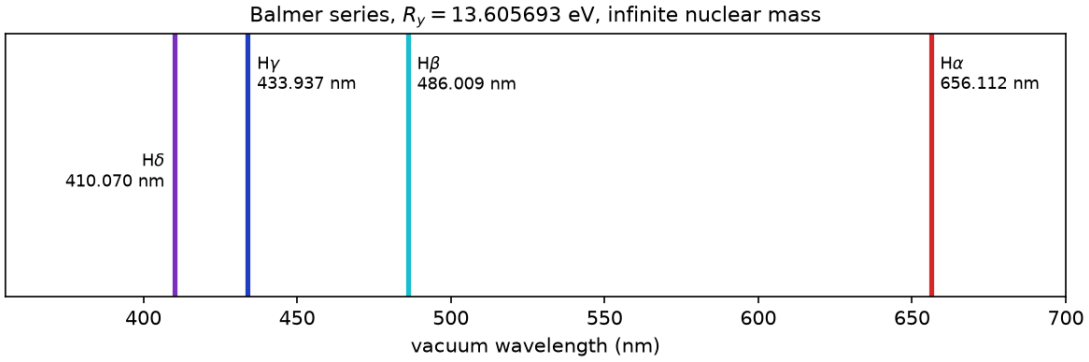


Fig. 3 — The Balmer series, computed for this note: Hα 656.112 nm, Hβ 486.009 nm, Hγ 433.937 nm, Hδ 410.070 nm, from $R_y = 13.605693$ eV and infinite nuclear mass (labels to three decimals, placed so that they do not overlap).

5.2 Error of these values: against Schrödinger and against experiment

Two different comparisons must be distinguished. (a) Against the Schrödinger equation with the same R_y, the lines of Table 3 differ only by the effect of the tail T on the levels. For the reading T' this is a relative level shift of order 10^{-126} for the $1s$ level (Section 4.6); for the printed form no unit-independent number exists. (b) Against measured wavelengths, the dominant systematic difference is the finite nuclear mass: with the reduced mass $m_r = m_e/(1 + m_e/m_p)$, $m_e/m_p = 9.1093837015 \times 10^{-31}/1.67262192 \times 10^{-27} = 5.44617 \times 10^{-4}$ [M], energies are multiplied by $1/(1 + m_e/m_p) = 0.999456$ and wavelengths by $1 + m_e/m_p$; for Hα this gives $656.112 \times 1.000544617 = 656.470$ nm [M]. This relative difference (5.4×10^{-4}) exceeds by a very large factor the $1s$ shift under T' of Section 4.6. Precision against experiment is limited by nuclear mass and by fine structure, not by the tail (E9). Measured wavelengths are not tabulated in this note.

6. Spin: the Pauli extension

The scalar equation does not describe spin. In this section the radial function after separation, $u = rg$ (Section 4.1), is promoted to a two-component spinor $\Psi = u(r)\chi_s\tau(t)$, $\chi_s^\dagger\chi_s = 1$, which corresponds to $f = g\tau$ of Section 3.4 with $u = rg$; the total Hamiltonian is $\hat{H} = \hat{H}_{\mathrm{ODE}} + \hat{H}_{LS} + \hat{H}_{\mathrm{Darwin}} + \hat{H}_{\mathrm{rel}} + \hat{H}_{HF}$ [P]. The terms are the standard ones [E] [1, 2].

6.1 Spin-orbit coupling

$$\hat{H}_{LS} = \xi(r)\,\vec{L}\cdot\vec{S}, \quad \xi(r) = \frac{1}{2m_e^2c^2}\frac{1}{r}\frac{dV}{dr} = \frac{\hbar\alpha}{2m_e^2c}\frac{1}{r^3} \text{ (Coulomb)}, \quad \langle\vec{L}\cdot\vec{S}\rangle$$

$$= \frac{\hbar^2}{2}[j(j+1) - l(l+1) - s(s+1)]. \qquad (19)$$

Dimensions: $\xi\,\vec{L}\cdot\vec{S}$ has dimension of energy [M].

6.2 Darwin term (s states)

$$\hat{H}_{\text{Darwin}} = \frac{\hbar^2}{8m_e^2c^2}\nabla^2 V = \frac{\pi\hbar^3\alpha}{2m_e^2 c}\,\delta^3(\vec{r}). \qquad (20)$$

With $\nabla^2 V = e^2\delta^3(\vec{r})/\varepsilon_0$ and $e^2/(4\pi\varepsilon_0) = \hbar c\alpha$ one obtains $\hbar^3$ [M] (E16).

6.3 Relativistic kinetic correction

$$\hat{H}_{\text{rel}} = -\frac{\hat{p}^4}{8m_e^3c^2}. \qquad (21)$$

6.4 Radial equation with spin-orbit coupling

Adding the spin-orbit term to (12) gives

$$u'' + u\left[\frac{2m_eE}{\hbar^2} + T(r) + \frac{2}{a_0 r} - \frac{l(l+1)}{r^2} + \frac{2m_e}{\hbar^2}\xi(r)\langle\vec{L}\cdot\vec{S}\rangle\right] = 0. \qquad (22)$$

Equation (22) contains the spin-orbit term only; it omits the Fermi term of (7) and, like (7), contains neither the Darwin term nor the p^4 correction (E21). The fine-structure formula of Section 7 is the standard result of first-order perturbation theory with all three terms [E] and is not derived from (22) alone.

7. Fine structure

To first order in α^2 the three terms of Section 6 give the Dirac formula [E] [1]

$$E_{n,j} = -\frac{R_y}{n^2} + \frac{R_y\alpha^2}{n^4}\left(\frac{3}{4} - \frac{n}{j+1/2}\right) + O(\alpha^4). \qquad (23)$$

7.1 Scale

$\alpha^2 = 5.325135\times10^{-5}$ and $\alpha^2 R_y = 5.325135\times10^{-5}\times 13.605693\ \text{eV} = 7.24522\times10^{-4}\ \text{eV} = 724.52\ \mu\text{eV}$ [M].

7.2 Splittings

$\Delta_{fs}(n,j) = (\alpha^2R_y/n^4)(3/4 - n/(j+1/2))$, $E_{n,j} = -R_y/n^2 + \Delta_{fs}$ [M].

Table 4 — Fine-structure shifts and levels.

n	l	j	α^2R_y/n^4 (μeV)	$3/4 - n/(j+1/2)$	Δ_{fs} (μeV)	$E_{n,j}$ (eV)
1	0	1/2	724.52	$-1/4$	-181.13	-13.605874
2	0	1/2	45.283	$-5/4$	-56.60	-3.4014799
2	1	1/2	45.283	$-5/4$	-56.60	-3.4014799
2	1	3/2	45.283	$-1/4$	-11.32	-3.4014346
3	0	1/2	8.9447	$-9/4$	-20.13	-1.5117638
3	1	1/2	8.9447	$-9/4$	-20.13	-1.5117638
3	1	3/2	8.9447	$-3/4$	-6.71	-1.5117504
3	2	3/2	8.9447	$-3/4$	-6.71	-1.5117504
3	2	5/2	8.9447	$-1/4$	-2.24	-1.5117459
4	3	5/2	2.8302	$-7/12$	-1.65	-0.8503575
4	3	7/2	2.8302	$-1/4$	-0.71	-0.8503565
5	4	7/2	1.1592	$-1/2$	-0.58	-0.5442283
5	4	9/2	1.1592	$-1/4$	-0.29	-0.5442280

Levels with the same n and j are degenerate in (23); the $2s_{1/2}$–$2p_{1/2}$ degeneracy is lifted only by the Lamb shift (Section 8) (E15).

7.3 The $2p_{3/2} - 2p_{1/2}$ splitting

$$E(2p_{3/2}) - E(2p_{1/2}) = \frac{R_y\alpha^2}{16} = \frac{13.605693\times5.3251354\times10^{-5}}{16}\ \text{eV} = 4.528264\times10^{-5}\ \text{eV} = 45.28\ \mu\text{eV}$$
$$= \frac{4.528264\times10^{-5}\ \text{eV}}{4.135668\times10^{-15}\ \text{eV s}} = 1.094929\times10^{10}\ \text{Hz} = 10949.29\ \text{MHz}. \qquad (24)$$

The measured splitting is obtained from two measured intervals [M]: the $2S_{1/2}$–$2P_{3/2}$ interval 9911.200(12) MHz [8] plus the Lamb shift $2S_{1/2}$–$2P_{1/2}$ 1057.845(9) MHz [7], giving $9911.200 + 1057.845 = 10969.045$ MHz. The difference from (24) is $(10969.045 - 10949.29)/10969.045 = 1.8\times10^{-3}$, i.e. 0.18% [M]; it is attributed to radiative (QED) corrections not contained in (23) [E] [1].

8. The Lamb shift

The splitting between $2s_{1/2}$ and $2p_{1/2}$ is measured as 1057.845(9) MHz [7] [E], which equals $h\times1.057845\times10^9$ Hz $= 4.135668\times10^{-15}$ eV s $\times\ 1.057845\times10^9\ \text{s}^{-1} = 4.375\ \mu$eV [M]. The effect was first observed by Lamb and Retherford [3]; its explanation lies in radiative corrections of quantum electrodynamics (self-energy and vacuum polarisation) [E] [1]. It is **not derivable** from the radial equations (12), (22): these have no quantised electromagnetic field. The leading-logarithm expression for an ns level is [E] [1]

$$\Delta E_{\text{Lamb}}(ns) \simeq \frac{4\alpha^5}{3\pi n^3}\,m_ec^2\ln\frac{1}{\alpha^2} + (\text{further terms}), \qquad (25)$$

and is not evaluated here. In this note the Lamb shift is inserted from measurement and is not predicted. Its derivation within TauTimp would require quantising the photon field in the undulatory formalism; this is an open problem.

9. Hyperfine structure: the 21 cm line

The line was first observed in galactic radiation by Ewen and Purcell [4]. The Fermi contact interaction between electron and proton spins is, with $g_e \simeq 2$ [E] [1, 2],

$$\hat{H}_{HF} = \frac{4\mu_0}{3\hbar^2}\, g_p \mu_B \mu_N\, \vec{S}_e \cdot \vec{I}_p\, \delta^3(\vec{r}). \qquad (26)$$

For the $1s$ state, $|\psi_{1s}(0)|^2 = 1/(\pi a_0^3)$, and with $\vec{F} = \vec{S}_e + \vec{I}_p$,

$$\langle \vec{S}_e \cdot \vec{I}_p \rangle = \frac{\hbar^2}{2}[F(F+1) - S(S+1) - I(I+1)] = \begin{cases} +\hbar^2/4 & F = 1 \\ -3\hbar^2/4 & F = 0 \end{cases} \qquad (27)$$

so the $F = 1$ to $F = 0$ splitting is $\hbar^2 \times \frac{4\mu_0}{3\hbar^2} g_p \mu_B \mu_N \frac{1}{\pi a_0^3}$:

$$\Delta E_{HF} = \frac{4\mu_0}{3} \frac{g_p \mu_B \mu_N}{\pi a_0^3}. \qquad (28)$$

Calculation [M]: $\mu_B \mu_N = 4.6841 \times 10^{-50}$ J²T⁻²; $\times\, g_p = 2.6164 \times 10^{-49}$; $\times\, 4\mu_0/3 = 1.67552 \times 10^{-6}$ gives 4.3839×10^{-55}; $\pi a_0^3 = 4.6554 \times 10^{-31}$ m³; so $\Delta E_{HF} = 9.417 \times 10^{-25}$ J $= 5.8774$ μeV, corresponding to $\Delta E_{HF}/h = 1421.160$ MHz. The same value follows from $4/3\, \alpha^4 g_p (m_e/m_p)\, m_e c^2 = 4/3 \times 2.83571 \times 10^{-9} \times 5.5856947 \times 5.44617 \times 10^{-4} \times 510998.95$ eV $= 5.8774$ μeV [M]; the two forms are identical because $|\psi_{1s}(0)|^2 \mu_0 \mu_B \mu_N$ reduces to $\alpha^4 m_e c^2 (m_e/m_p)$ times a constant [E].

The measured frequency is 1420.405752 MHz [9] (= 5.8743 μeV). Formula (28) exceeds it by $(1421.160 - 1420.406)/1420.406 = 5.3 \times 10^{-4}$ (0.053%) [M]. This difference is attributed to corrections not included in the contact formula: the reduced mass in $|\psi(0)|^2$, the anomalous magnetic moment of the electron ($g_e \neq 2$), and QED and proton-structure corrections [E].

The hyperfine term enters (7) as a contact term $V_{HF}(r)\langle \vec{S}_e \cdot \vec{I}_p \rangle$; as for fine structure, the value above is the standard first-order result [E] and (7) adds no new prediction.

10. The normal Zeeman effect

In a field $\vec{B} = B\hat{z}$, if spin is neglected, each level (n, l) splits into $2l + 1$ sublevels [E]

$$\Delta E_{m_l} = m_l \mu_B B, \qquad m_l = -l, \ldots, +l. \qquad (29)$$

l	$\lvert\vec{\mu}_L\rvert/\mu_B = \sqrt{l(l+1)}$	Sublevels
1 (p)	$\sqrt{2} = 1.414$	3
2 (d)	$\sqrt{6} = 2.449$	5
3 (f)	$\sqrt{12} = 3.464$	7
4 (g)	$\sqrt{20} = 4.472$	9

At $B = 1$ T: $\mu_B B = 9.2740101 \times 10^{-24}$ J$/1.602176634 \times 10^{-19}$ J/eV $= 5.7884 \times 10^{-5}$ eV $= 57.88$ μeV, and $\mu_B B/h = 9.2740101 \times 10^{-24}/6.62607015 \times 10^{-34} = 1.3996 \times 10^{10}$ Hz $= 13.996$ GHz [M]. The electron has spin 1/2 with $g_s \simeq 2$; (29) is the idealisation without spin, and the complete Zeeman pattern of hydrogen (anomalous at weak field) is not treated here [E] (E23).

11. Hydrogen orbitals

The eigenfunctions of (8) are $\psi_{nlm} = R_{nl}(r) Y_l^m(\theta, \phi)$ [E] [2]. In units $a_0 = 1$:

$$R_{10} = 2e^{-r}, \quad R_{20} = 1/2\sqrt{2}\,(2 - r)e^{-r/2}, \quad R_{21} = 1/2\sqrt{6}\; r\, e^{-r/2},$$
$$R_{30} = 2/81\sqrt{3}\,(27 - 18r + 2r^2)e^{-r/3}, \quad R_{31} = 4/81\sqrt{6}\; r(6 - r)e^{-r/3}, \quad R_{32} = 4/81\sqrt{30}\; r^2 e^{-r/3}. \qquad (30)$$

For $l = n - 1$ the function has no radial node and $w_{nl} = r^2 R_{nl}^2 \propto r^{2n} e^{-2r/n}$ is maximal where $d\ln w_{nl}/dr = 2n/r - 2/n = 0$, i.e. at $r = n^2 a_0$ [M]: $1s$ at $1\, a_0$, $2p$ at $4\, a_0$, $3d$ at $9\, a_0$ (Fig. 4).

Relation to (12). For the reading T' the first-order shift of a bound level is the finite expectation value $\langle T' \rangle$ (Section 4.6) [M, reading T']. The form of the eigenfunctions of (12) on $[0, R]$ with $g(R) = 0$ is not established here: the tail grows as r^4 and is not uniformly small on $[0, R]$, so that statement is a hypothesis [H].

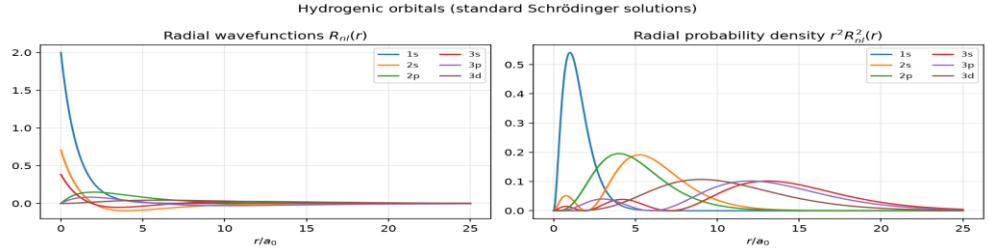


Fig. 4 — Hydrogenic orbitals (standard Schrödinger solutions), computed for this note, for the states 1s, 2s, 2p, 3s, 3p, 3d over $r/a_0 \in [0,25]$. Left panel: radial wavefunctions $R_{nl}(r)$. Right panel: radial probability density $r^2 R_{nl}^2(r)$. Normalisation checked: $\int_0^\infty r^2\, R_{nl}^2\, dr = 1$.

12. The probability-density wave [H]

This section is a hypothesis; its equations contain dimensional problems, which are marked.

12.1 Equation

For constant velocity $V = \alpha c = 2.18769 \times 10^6$ m/s (Bohr-orbit speed) the probability density $\varrho(r,t)$ is assumed [P] to obey $\frac{\partial^2 \varrho}{\partial t^2} - V^2 \frac{\partial^2 \varrho}{\partial r^2} = \frac{V^2}{R}\,\varrho, \qquad \varrho(R,0) = 0. \qquad (31)$

Dimensional note. V^2/R has dimension $\mathrm{m/s^2}$, while the left-hand side has dimension $\mathrm{s^{-2}}$ times that of ϱ; (31) is not dimensionally consistent as printed [M]. A consistent form would carry V^2/R^2 (or another length); the specification does not give it. Equation (31) is retained as a postulate with this reservation (E12). For the variant with V^2/R^2 the plane-wave dispersion relation would be $\omega^2 = V^2(k^2 - 1/R^2)$, with $1/R = 2.36 \times 10^{-23}$ $\mathrm{m^{-1}}$ [M, conditional].

12.2 The wavenumber

For $\varrho \propto \sin(k_e r)\cos(\omega_e t)$ with the wave velocity $V = \alpha c$ and momentum $m_e V$, the de Broglie-type wavenumber is [H]

$$k_e = \frac{m_e V}{\hbar} = \frac{m_e \alpha c}{\hbar} = \frac{\alpha}{\bar{\lambda}_C} = \frac{1}{a_0} = 1.8897 \times 10^{10}\ \mathrm{m^{-1}}, \qquad \omega_e = V k_e = \frac{\alpha c}{a_0} = 4.134 \times 10^{16}\ \mathrm{rad/s}. \qquad (32)$$

Calculation: $\alpha \times 2.589605 \times 10^{12} = 1.88972 \times 10^{10}$; $2.18769 \times 10^6 \times 1.88972 \times 10^{10} = 4.1341 \times 10^{16}$ (E11).

12.3 Zeros

The zeros of $\sin(k_e r)$ lie at $r_b = b\pi/k_e = b\pi a_0$, $b = 1,2,\ldots$, spacing $\pi a_0 = 1.6624 \times 10^{-10}$ m [M]. (Even with $k = 1/\bar{\lambda}_C$ the spacing would be $\pi\bar{\lambda}_C = 1.2132 \times 10^{-12}$ m, not $\bar{\lambda}_C$.) The equality $\alpha a_0 = \bar{\lambda}_C = 3.8616 \times 10^{-13}$ m is an identity of the constants [M]; it does not follow from (31). This calculation does not show a coincidence of the zeros with $\bar{\lambda}_C$; the length αa_0 is named here only as a scale of interest [H].

12.4 The Gaussian envelope

With $\varrho(r,0) = \exp[-(r - r_B)^2/(2\sigma^2)]$ and $\partial_t \varrho(r,0) = 0$, and neglecting the right-hand side of (31), d'Alembert's solution is $1/2\,[G(r - Vt) + G(r + Vt)]$, two Gaussians travelling in opposite directions [E]. A factor $\sin(k_e r)\cos(\omega_e t_0)$ multiplying them does not follow from the solution; it is added by hand. The product form is a hypothesis [H].

13. Hierarchy of scales

Table 5 — Hierarchy.

Scale	Effect	Value
R_y	Bohr (coarse) structure	13.6057 eV
$\alpha^2 R_y$	fine structure	724.52 μeV
$\sim \alpha^3 R_y$ (up to logarithm and numerical factor)	Lamb shift	4.375 μeV (measured, inserted)
$\sim \alpha^4 (m_e/m_p)\, m_e c^2$	hyperfine (21 cm)	5.8774 μeV (formula), 5.8743 μeV (measured)
$\mu_B B$	normal Zeeman at 1 T	57.88 μeV per unit m_l
tail/Coulomb, reading T'	cosmological tail (Section 4.6)	$\leq 1.9 \times 10^{-121}$ for $r \leq 21\, a_0$ (dimensionless)
tail/Coulomb, printed T	cosmological tail (Section 4.6)	unit $\mathrm{m^2}$, unit-dependent; $\leq 2.2 \times 10^{-51}$ after division by L^2 with $L \geq \bar{\lambda}_C$ [conditional]

14. Discussion

What the note shows. (a) The equation of the specification, read literally, differs from the Schrödinger equation at small r (Section 4.3) [M]. (b) After three modifications it becomes the Schrödinger equation plus the tail of the specification (Section 4.4) [M]; this is an identity of construction. (c) For the dimensionally consistent reading T' the ratio of the tail to the Coulomb term is below 1.9×10^{-121} at $r \leq 21\, a_0$, and the first-order shift of the $1s$ level is 2.8×10^{-125} eV [M]; $\langle T' \rangle$ for other states scales as $\langle r^4 \rangle$ and is not tabulated here. For the printed form the ratio has the unit $\mathrm{m^2}$ and no unit-independent statement can be made, apart from the conditional bound of Section 4.6. (d) The standard results for hydrogen in Sections 5–11 are those of Schrödinger, Pauli and Dirac theory [E], recomputed here from the constants of Table 1.

What it does not show. The note does not show that the modifications (i)–(iv) follow from TauTimp; they have the effect of recovering the Schrödinger operator. It does not show that TauTimp explains any hydrogen result: after the modifications the atomic results are those of Schrödinger theory, and the tail adds a term whose $1s$ shift under T' is 2.8×10^{-125} eV. It does not show that the modifications are unique. The standard results of Sections 5–11 are recovered by (12) only after modifications that make its radial operator the Schrödinger operator. A discriminating test of TauTimp would have to be made where the tail is not negligible, at r approaching R; it is not made here.

Limitations

1. **Construction.** The agreement with the Schrödinger equation is by construction (Section 4.4); it is not an independent verification.
2. **Underived steps.** The passage from the derivation (6) to the specification form (10), the form of the tail, the separation step of Section 3.3, modifications (i)–(iv), and the spin extension (7) are postulates [P] or not shown.
3. **Dimensions.** The tail T as printed is m^0 in a bracket of m^{-2} (Section 4.6), the centrifugal term $l(l+1)/R$ of (10) is m^{-1}, and the right-hand side of (31) is dimensionally inconsistent. The values of Table 2 for the printed tail are SI numbers of a quantity with unit m^2 and depend on the unit of length.
4. **Lamb shift** is inserted from measurement, not derived (Section 8).
5. **Relativistic extension.** Equation (7) contains the spin-orbit term and the Fermi term V_{HF}, and (22) contains the spin-orbit term only; neither contains the Darwin term or the p^4 correction, which enter only by first-order perturbation theory. A full Dirac-type TauTimp equation is not treated.
6. **Hyperfine** formula (28) is the contact term with $g_e \simeq 2$ and infinite proton mass; it differs from the measured value by 0.053%.
7. **Zeeman effect** is treated without spin.
8. **Eigenfunctions of (12)** on $[0, R]$ with $g(R) = 0$ are not studied (Section 11) [H].
9. **Section 12** is a hypothesis with unresolved dimensional problems.
10. **Not tested.** No test of TauTimp at $r \sim R$ is made.

Conclusions

1. The electron equation of the working specification, taken literally, does not agree with the Schrödinger equation at small r: the coefficients of $r^2(\mathcal{L}_S - \mathcal{L}_{\mathrm{raw}})$ are 0, $-g_0 l(l+1)$, $-g_1 l(l+1) + 4g_0/a_0$ at orders r^0, r^1, r^2 [M].
2. Three modifications (Coulomb sign, bare energy term, localised centrifugal term) turn it into the Schrödinger operator plus the tail T of the specification [P]; the equation (12) equals (8) plus $T\,u$ identically [M]. The Taylor coincidence through r^6 (convention of Section 4.2; equivalently through r^4 for the unmultiplied expressions) is a consequence of this construction, not an independent test of TauTimp [M].
3. If the derivation (6) is the starting point, the factor r/R of (9) must also be removed (modification (iv)) and the tail added as a postulate; (9) itself is not the Schrödinger equation [M].
4. The printed tail has dimension m^0 in a bracket of m^{-2}; its ratio to the Coulomb term has the unit m^2 and is unit-dependent (3.35×10^{-76} m^2 $= 3.4 \times 10^{-46}$ fm^2 at 21 a_0); after division by L^2 it is at most 2.2×10^{-51} for $L \geq \bar{\lambda}_C$ [M, conditional]. For the dimensionally consistent reading $T' = (r/R)^4(2m_eE/\hbar^2 - \kappa^2)$ the ratio is 1.5×10^{-131}, 4.6×10^{-128}, 1.9×10^{-121} at 0.2, 1, 21 a_0 and the first-order shift of the $1s$ level is 2.8×10^{-125} eV [M]; the shifts of other states scale as $\langle r^4 \rangle$ and are not tabulated here. Only for the reading T' is the ratio shown to be negligible at $r \leq 21\,a_0$.
5. The Bohr spectrum and the ten spectral lines of Table 3 are standard results recomputed with $R_y = 13.605693$ eV, $hc = 1239.84193$ eV·nm and infinite nuclear mass; with the reduced mass Hα is 656.470 nm [M]. The difference from experiment at the level of 5.4×10^{-4} is the nuclear mass and not the tail [M].
6. The fine-structure scale is $\alpha^2 R_y = 724.52$ μeV; the $2p_{3/2} - 2p_{1/2}$ splitting is 45.28 μeV $= 10949.29$ MHz, 0.18% below the measured 10969.045 MHz [7, 8] [E, M].
7. The Lamb shift (4.375 μeV) is outside the equations of this note and is inserted from measurement [E].
8. The hyperfine splitting from the contact term (28) is 5.8774 μeV $= 1421.160$ MHz, 0.053% above the measured 1420.405752 MHz [9]; the prefactor is $4\mu_0/(3\hbar^2)$ [M].
9. The normal Zeeman energy at 1 T is 57.88 μeV $= 13.996$ GHz per unit m_l, spin neglected; the hydrogen orbitals and the maxima of w_{nl} at $n^2 a_0$ for $l = n - 1$ are standard results [E, M].
10. The probability-density wave of Section 12 is a hypothesis [H]. Its wavenumber is $k_e = 1/a_0 = 1.8897 \times 10^{10}$ m^{-1}, its angular frequency $\omega_e = 4.134 \times 10^{16}$ rad/s, and its zeros lie at $b\pi a_0$, $b = 1,2,\ldots$ [M]; the length $\alpha a_0 = \bar{\lambda}_C$ is an identity of the constants [M].
11. At atomic scale the modified TauTimp equation is the Schrödinger equation plus the tail of the specification. Under the reading T' the ratio of the tail to the Coulomb term is $\leq 1.9 \times 10^{-121}$ for $r \leq 21\,a_0$ and the $1s$ shift is 2.8×10^{-125} eV [M]; the discriminating test lies at $r \sim R$ and is not made here [H].

Appendix A. The algebra of Sections 4.3–4.5

A.1 Equation (13). $\mathcal{L}_{\mathrm{mod}}[u] - \mathcal{L}_S[u] = u\,(B_{\mathrm{mod}} - B_S) = u\,T(r)$ because the bracket of (12) is $B_S + T$ [M].

A.2 Coefficients (11), by hand. $r^2 u\,(B_S - B_{\mathrm{raw}})$ with $u = g_0 r + g_1 r^2 + g_2 r^3 + \cdots$: the term $-l(l+1)/r^2$ gives $-l(l+1)u = -l(l+1)(g_0 r + g_1 r^2 + \cdots)$; the term $4/(a_0 r)$ gives $4ru/a_0 = 4g_0 r^2/a_0 + \cdots$; the term $2m_eE/\hbar^2$ gives $r^2 u \cdot 2m_eE/\hbar^2$, of order r^3; the terms $l(l+1)/R$ and T give orders $\geq r^3$ and r^7. Collecting r^1 and r^2 gives (11) [M].

A.3 Equation (14). $T\,u = r^4(2m_eE/\hbar^2 - \kappa^2)R^{-2}(g_0 r + \cdots)$, so $\mathcal{L}_S - \mathcal{L}_{\mathrm{mod}} = -Tu$ has leading term $g_0(\kappa^2 - 2m_eE/\hbar^2)r^5/R^2 = g_0(\hbar^2\kappa^2 - 2m_eE)r^5/(R^2\hbar^2)$ [M].

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Appendix B. Changes relative to the unpublished draft of 26 August 2026

"Printed" is the text of the earlier version; "Recomputed" is the result in this version; "Disposition" states what is done.

Item	Printed	Recomputed	Disposition
E1. Abstract, §4.2–4.3, §14, Concl. 3 and 11	Taylor coincidence through r^4 is a "rigorous equivalence", "exactly reproduces", "algebraically exact extension"	$\mathcal{L}_{\mathrm{mod}} = \mathcal{L}_S + T\,u$ identically (13); coincidence follows from construction	Withdrawn; replaced by §4.4 and Conclusions 2, 11
E2. §3.2 → §4.2	(5)→(6) presented as the three modifications	(9): $u'' + (r/R)uB_S = 0$; the factor r/R is removed by a fourth modification; relation of (9) to (10) not derived	Declared: modification (iv); (10) kept as specification form [P] (§4.1, §4.4)
E3. §2.3, eq. (1)	"exactly as specified" with +Coulomb, −centrifugal	Specification: −Coulomb, +centrifugal, no $1/r$ in the Coulomb term	Specification form shown as (1); change of signs stated (§3.1)
E4. §0, §4.2, §4.4, Fig. 1	$k^2 = 2m_e c^2/\hbar^2$; residuals 2.88×10^{-56}, 9.01×10^{-53}, 3.68×10^{-46}	$\kappa^2 = 2/\bar{\lambda}_C^2 = 1.34121 \times 10^{25}$ m^{-2}; ratio $= r^5\kappa^2 a_0/(2R^2) = 2.63 \times 10^{-86}$, 8.21×10^{-83}, 3.35×10^{-76} (unit m^2)	Replaced; Fig. 1 regenerated; dimensional status added (§4.6)
E5. §4.1	r^0 coefficient $-g_1 l(l+1) + 4g_0/a_0$; title "divergence from r^0"	r^0: 0; r^1: $-g_0 l(l+1)$; r^2: $-g_1 l(l+1) + 4g_0/a_0$	Corrected (§4.3, eq. (11)); Concl. 2 ("r^0–r^1") corrected
E6. §4.3	first deviation at r^5, convention not stated	r^5 for unmultiplied expressions; r^7 after multiplication by r^2	Convention stated (§4.5)
E7. §4.5	$1/R^2 \sim 10^{-90}$ m^{-2}	$1/R^2 = 5.574 \times 10^{-46}$ m^{-2}	Corrected
E8. §2.2	$R/r^2 \sim 10^{44}$, "exactly the factor needed"	$R/a_0^2 = 1.51 \times 10^{43}$ m^{-1} (dimensional); $R/a_0 = 8.00 \times 10^{32}$ (dimensionless); "needed" not shown	Statement removed; $(R/r)\Delta f$ form given (§3.2)
E9. §5.1	Lyα 121.518, Lyβ 102.531, Lyγ 97.205, Hα 656.196, Hβ 486.071, Hγ 433.992, Hδ 410.119, Paα 1874.845, Paβ 1281.633, Brα 0.3062 eV / 4050.0; "relative error $\lesssim 10^{-46}$,	Table 3 (121.502, 102.518, 97.202, 656.112, 486.009, 433.937, 410.070, 1874.606, 1281.469, 0.3061 eV / 4050.08); printed values deviate by up to 1.3×10^{-4} (Hα: 0.084/656.112); finite	Replaced; statement on error withdrawn and reformulated (§5.2)

Item	Printed	Recomputed	Disposition
	exact to any experimental precision"	nuclear mass gives Hα 656.470 nm (5.4×10^{-4})	
E10. §2.5, §9	$V_{HF} = 8\mu_0 g_p \mu_B \mu_N / (3\hbar^2)\, \delta^3$; gives 2842 MHz; formula "$= h \times$ 1420.40575 MHz"	$4\mu_0/(3\hbar^2)$; 5.8774 μeV = 1421.160 MHz; measured 1420.405752 MHz, difference 0.053%	Corrected; difference declared (§9)
E11. §12.3–12.4, Concl. 10	$k_e = 1/(\alpha a_0) = 137/a_0 = 2.5896 \times 10^{12}$ m^{-1}; $\omega_e = 5.665 \times 10^{18}$; zeros at $b\, r_B/137$; coincidence with $\bar{\lambda}_C$	$k_e = m_e \alpha c/\hbar = 1/a_0 = 1.8897 \times 10^{10}$ m^{-1}; $\omega_e = 4.134 \times 10^{16}$ rad/s; zeros at $b\pi a_0 = b \times 1.6624 \times 10^{-10}$ m; the printed 2.5896×10^{12} m^{-1} is $m_e c/\hbar = 1/\bar{\lambda}_C$ and 5.665×10^{18} s^{-1} = $V/\bar{\lambda}_C = 2.18769 \times 10^6 \times 2.589605 \times 10^{12} = 5.6653 \times 10^{18}$ mixes $V = \alpha c$ with $k = 1/\bar{\lambda}_C$	Corrected; coincidence claim withdrawn (§12.2–12.3)
E12. §12.1–12.2	$P_{tt} - V^2 P_{rr} = (V^2/R)P$; "tachyonic Klein–Gordon, $m^2c^4 = -\hbar^2 V^2/R$"; $k \gg 1/\sqrt{R}$	V^2/R in m/s^2, not s^{-2}; $1/\sqrt{R}$ in m$^{-1/2}$	Kept as postulate [P] with the dimensional note; identification and condition removed (§12.1)
E13. §12.5	"Gaussian envelope confirmed"	d'Alembert gives two travelling Gaussians; factor $\sin(k_e r)\cos(\omega_e t_0)$ added by hand	Relabelled [H]; "confirmed" withdrawn (§12.4)
E14. §2.4	"absorbing ω^2/V^2 … $V^2 = 2E/m_e$"	$E < 0 \Rightarrow V^2 < 0$; with $E = \hbar\omega$, $\omega^2/V^2 = 1/4$ $(2m_e E/\hbar^2)$	Step marked as not shown (§3.3)
E15. §7.2	Δ_{fs} $-1.42, -0.57$ $(n = 4)$, $-0.47, -0.23$ $(n = 5)$; $E_{n,j}$ column, e.g. -0.8503839 eV $(n = 4, j = 5/2)$	$-1.65, -0.71$; $-0.58, -0.29$; $E_{n,j}$ column of Table 4 (e.g. -0.8503575 eV)	Replaced
E16. §6.2	$\widehat{H}_{\text{Darwin}} = \pi\hbar^2 \alpha\, \delta^3 /(2m_e^2 c)$	dimension s^{-1}; correct: $\pi\hbar^3\alpha/(2m_e^2 c)\, \delta^3$	Corrected
E17. §3.2, (5′)	$l(l+1)/R$ in the bracket	dimension m^{-1} (other terms m^{-2})	Noted (§4.1); removed by modification (iii)
E18. §4.2, §4.4, Fig. 1 caption	tail $r^4(2m_e E/\hbar^2 - k^2)/R^2$ in formulas but "$(r/R)^4$" in text and caption; tail "undetectable in both readings"	printed tail has dimension m^0; ratio to Coulomb is in m^2 and unit-dependent (3.4×10^{-46} fm^2 at 21 a_0); reading $(r/R)^4$ gives $4.6 \times 10^{-128}\, x^5$	Both shown, status stated; negligibility claimed only for T' (§4.6, Table 2)
E19. §4.4	"more than 30 orders of magnitude" below double precision	1.87×10^{-121} (T'): 105 orders; 3.35×10^{-76} m^2 (printed, SI value only): 59 orders	Corrected (§4.6)
E20. §0, (2.2)	k used for $k^2(r) = m_e^2 V^2/\hbar^2$ and for $k^2 = 2m_e c^2/\hbar^2$; also T for tail and time factor, u for $u(t)$ and $u(r)$, P for density, μ for reduced mass	two different quantities each	Renamed $q(r)$, κ, $\tau(t)$, ϱ, m_r
E21. §6.4, §2.5	(7), (3) "complete equation"	(7) contains the spin-orbit term and the Fermi term V_{HF}; (22) contains the spin-orbit term only; neither contains	Stated (§3.4, §6.4)

Item	Printed	Recomputed	Disposition
		the Darwin term or the p^4 correction	
E22. §13	Lamb scale "$\alpha^3 \ln \alpha$" without units	$\sim \alpha^3 R_y$ up to logarithm and factor	Reworded
E23. §10	normal Zeeman effect presented without qualification; $\simeq 14.00$ GHz	spin neglected; 13.996 GHz	Qualified; value corrected
E24. Header, App. A.2	list of deliverables including a Maple transcript .mpl	the transcript referred to is not part of this document	Removed
E25. §4.7	hierarchy "(1) centrifugal $\gg$ (2) Coulomb $\gg$ (3) energy $\gg$ (4) tail" at $r \sim a_0$, and "hence (i)–(iii) suffice"	at $r = a_0$, $l = 1$: centrifugal $=$ Coulomb $= 2/a_0^2$; for $l = 0$ no centrifugal term; at $21\ a_0$ the energy term ($n \leq 3$) exceeds the Coulomb term	Removed; replaced by the direct comparison of §4.7
E26. §11	radial solutions of (6) are the Schrödinger functions "up to the residual"	only the first-order shift $\langle T' \rangle$ is established; eigenfunctions on $[0, R]$ with $g(R) = 0$ not studied	Relabelled [H] (§11)
E27. §7.3	measured $2p_{3/2} - 2p_{1/2}$ splitting 10.969 GHz without source	$9911.200 + 1057.845 = 10969.045$ MHz from [8], [7]; difference from 10949.29 MHz is 0.18%	Sources and calculation added

Acknowledgement of tools

AI tools were used to recompute numerical values, check algebraic identities, generate the figures from the formulas stated, and edit the text.